



AWS Cloud Computing Concepts

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Cloud computing is the on-demand delivery of compute power, database storage,



applications,
and other IT resources through a cloud services
platform via the Internet with pay-as-you-go pricing.

Cloud computing provides a simple way to access
servers, storage, databases, and a broad set of
application services over the Internet.

The following introductory-level article covers some
key **AWS concepts** that relate to cloud computing:

- **What is Cloud Computing? Cloud vs Legacy IT**

The following articles provide some additional
information around basic computing concepts:

- **Cloud Computing Basics – Compute**
- **Cloud Computing Basics – Storage**
- **Cloud Computing Basics – Network**
- **Cloud Computing Basics – Serverless**

A cloud services platform such as Amazon Web
Services owns and maintains the network-
connected hardware required for these application
services, while you provision and use what you need
via a web application.

The 6 Advantages of Cloud

You must understand the following 6 advantages of
cloud:

1. Trade capital expense for variable expense.



2. Benefit from massive economies of scale.
3. Stop guessing about capacity.
4. Increase speed and agility.
5. Stop spending money running and maintaining data centers.
6. Go global in minutes.

Trade capital expense for variable expense

Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.

Benefit from massive economies of scale

By using cloud computing, you can achieve a lower variable cost than you can get on your own.

Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go price.

Stop guessing about capacity

Eliminate guessing on your infrastructure capacity needs. When you make a capacity decision prior to deploying an application, you often end up either sitting on expensive idle resources or dealing with limited capacity.

With cloud computing, these problems go away. You can access as much or as little capacity as you need and scale up and down as required with only a few minutes' notice.

Increase speed and agility

In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes.

This results in a dramatic increase in agility for the organization since the cost and time it takes to experiment and develop is significantly lower.

Stop spending money running

and maintaining data centers

Focus on projects that differentiate your business, not the infrastructure. Cloud computing lets you focus on your own customers, rather than on the heavy lifting of racking, stacking, and powering servers.

Go global in minutes

Easily deploy your application in multiple regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

Cloud Computing Models

There are 3 common types of cloud computing model that come up in the exam:

1. Infrastructure as a service (IaaS).
2. Platform as a service (PaaS).
3. Software as a service (SaaS).

Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) contains the basic building blocks for cloud IT and typically provide access to networking features, computers (virtual or on dedicated hardware), and data storage space.

IaaS provides you with the highest level of flexibility and management control over your IT resources and is very similar to the existing IT resources that many IT departments and developers are familiar with today.

Platform as a Service (PaaS)

Platform as a Service (PaaS) removes the need for your organization to manage the underlying infrastructure (usually hardware and operating systems) and allows you to focus on the deployment and management of your applications.

This helps you be more efficient as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.

Software as a Service (SaaS)

Software as a Service (SaaS) provides you with a completed product that is run and managed by the service provider. In most cases, people referring to Software as a Service are referring to end-user applications.

With a SaaS offering you do not have to think about how the service is maintained or how the underlying infrastructure is managed; you only need to think

about how you will use that piece of software.

A common example of a SaaS application is web-based email which you can use to send and receive email without having to manage feature additions to the email product or maintain the servers and operating systems that the email program is running on.

SaaS provides high availability, fault tolerance, scalability and elasticity.

The following article provides some additional information:

- **Cloud Computing Service Models – IaaS, PaaS, SaaS**

Types of Cloud Deployment

There are 3 common types of cloud deployment that come up in the exam:

1. Public Cloud – e.g. AWS, Microsoft Azure, Google Cloud Platform (GCP).
2. Hybrid Cloud – a mixture of public and private clouds.
3. Private Cloud (on-premises) – a cloud managed in your own data center, e.g. Hyper-V, OpenStack, VMware.

Public Cloud

A cloud-based application is fully deployed in the cloud and all parts of the application run in the cloud. Applications in the cloud have either been created in the cloud or have been migrated from an existing infrastructure to take advantage of the benefits of cloud computing.

Cloud-based applications can be built on low-level infrastructure pieces or can use higher level services that provide abstraction from the management, architecting, and scaling requirements of core infrastructure.

Hybrid

A hybrid deployment is a way to connect infrastructure and applications between cloud-based resources and existing resources that are not located in the cloud.

The most common method of hybrid deployment is between the cloud and existing on-premises infrastructure to extend, and grow, an organization's infrastructure into the cloud while connecting cloud resources to the internal system.



On-premises

The deployment of resources on-premises, using virtualization and resource management tools, is sometimes called the “private cloud.”

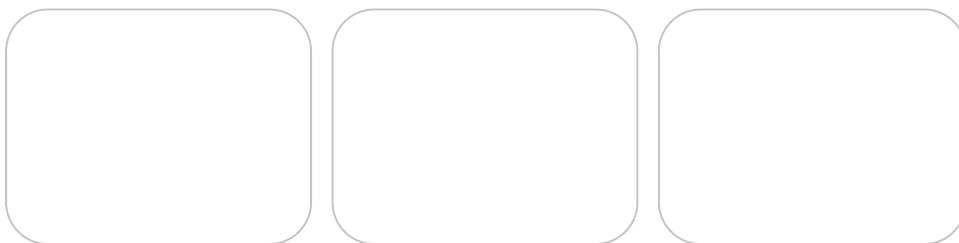
On-premises deployment doesn’t provide many of the benefits of cloud computing but is sometimes sought for its ability to provide dedicated resources.

In most cases this deployment model is the same as legacy IT infrastructure while using application management and virtualization technologies to try and increase resource utilization.

The following article provides some additional information:

- **[Cloud Computing Deployment Models – Public, Private & Hybrid](#)**

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